McGill initiative in Computational Medicine

McGill | CDSI Computational and Data Systems Institute | ISCD Institut en systèmes computationnels et de données

D2R DNA TO RNA | HeDS

Canadian Centre for Computational Genomics

CODE TO MANUSCRIPT BOOTCAMP · DAY 1

Data Visualization in Python

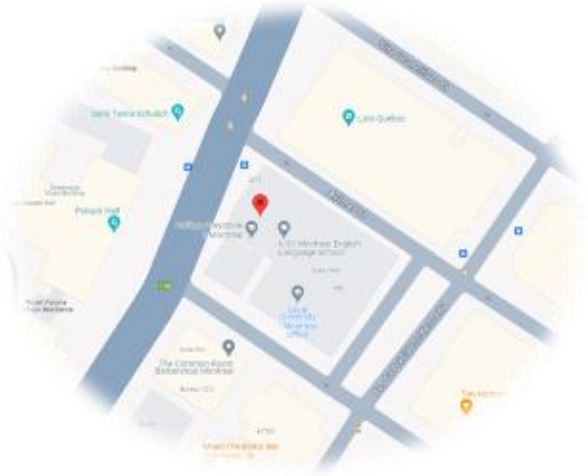
From RNA-seq counts to publication-ready figures with Matplotlib & Seaborn

Instructor: Zahra Tavakol Hamedani
Facilitator: Sameena Karsan
October 26, 2026 · McIntyre 1027

McGill | Quantitative Life Sciences | Sciences quantitatives du vivant

MiCM McGill initiative in Computational Medicine

QLS-MiCM mission statement: deliver quality workshops designed to help biomedical researchers develop the skills they need to succeed.



Location: 550 Sherbrooke Street, Montreal, Quebec

Scan the QR code to sign up for our **mailing list**

Contact: workshop-micm@mcgill.ca

McGill Quantitative Life Sciences | Sciences quantitatives du vivant

McGill initiative in Computational Medicine

Workshop Series

https://www.mcgill.ca/micm/training/workshops-series

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The Code to Manuscript bootcamp

Day 1
Mon Oct 26
Data visualization in Python
Matplotlib · Seaborn

Day 2
Tue Oct 27
Data visualization in R
ggplot2

Day 3
Wed Oct 28
Bring your own data
Python or R

Day 4
Thu Oct 29
Building figures with Inkscape
Inkscape

Day 5
Fri Oct 30
Writing with LaTeX
Overleaf

▲ TODAY

Today's goal: turn an RNA-seq dataset into a multi-panel, publication-ready figure in Python.

Today's output → tomorrow's input: Day 2 uses the **same dataset** in R; the vector PDF/SVG you export today is what you polish in **Inkscape** (Day 4) and place in **LaTeX** (Day 5).

Learning outcomes

In this 2-hour hands-on session we explore the Python visualization ecosystem: how Seaborn's high-level statistical plots are built on Matplotlib Figure and Axes objects, and how to combine both to produce clear, high-resolution figures from real biological data.

1. **Explain** the Figure / Axes model and use Matplotlib's object-oriented interface.
2. **Reshape** data into tidy format and use axes-level vs. figure-level Seaborn functions.
3. **Create** QC, distribution, scatter, volcano, heatmap and PCA / t-SNE / UMAP plots of RNA-seq data.
4. **Assemble** a multi-panel figure at journal dimensions with a consistent, colour-blind-safe style.
5. **Export** vector (PDF/SVG) and high-dpi raster files ready for Inkscape and LaTeX.

Outline

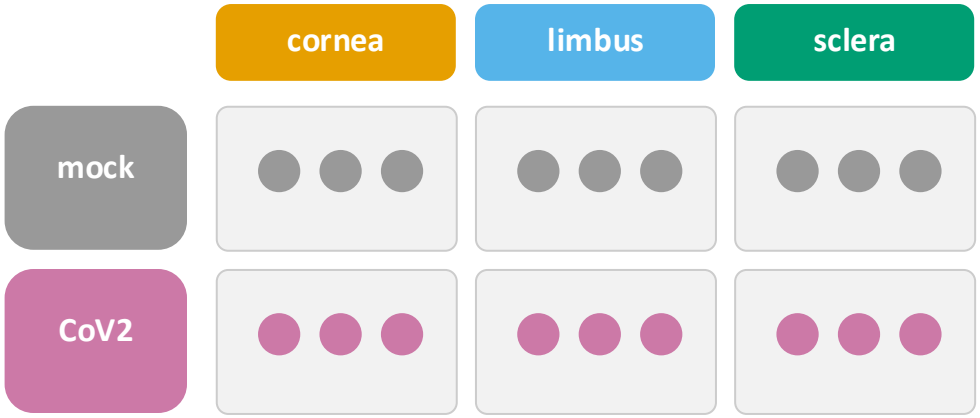
Module	Topic	Minutes
0	Setup & the dataset	10
1	Anatomy of a Matplotlib figure	15
2	Seaborn, tidy data & distributions	20
3	Relationships: scatter, MA & volcano plots	20
4	Heatmaps & clustered heatmaps	15
5	Dimensionality reduction: PCA, t-SNE, UMAP	15
6	Multi-panel figures & high-resolution export	20
	Wrap-up, checklist & feedback	5



Every module ends with a short exercise in the notebook.

Today's dataset: SARS-CoV-2 & the eye

Bulk RNA-seq of post-mortem human ocular tissue infected ex vivo with SARS-CoV-2 or mock — *Eriksen et al., Cell Stem Cell 2021* · **GEO GSE164073**



3 tissues × 2 conditions × 3 donors = **18 samples**

`Exercises/data/`

`sample_metadata.csv`
sample · tissue · condition

`GSE164073_Eye_count_matrix.tsv`
raw counts, genes × samples

`log2cpm_filtered.csv`
normalised expression

`de_cornea_CoV2_vs_mock.csv`
DE results (PyDESeq2)

Same data in **Day 2 (ggplot2)** and in the QLS-MiCM Differential Gene Expression workshop. Today we focus on ***showing*** data, not on the statistics behind it.

Interactive workshop — open the notebook!



github.com/QLS-MiCM/
DataVisualizationInPython

▶ Open in Colab

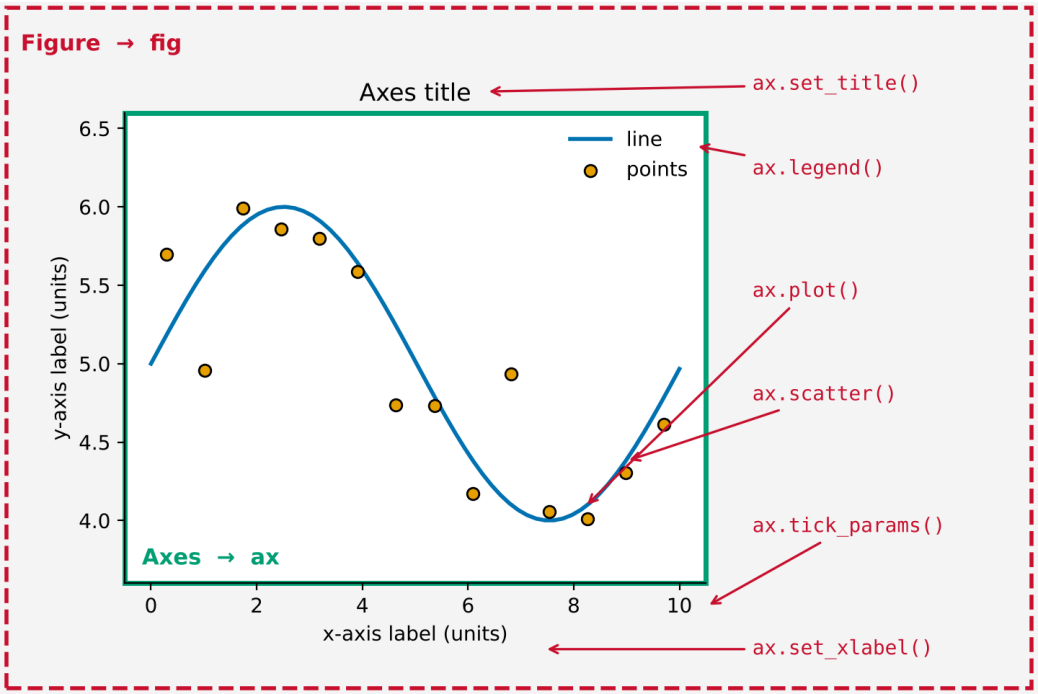
1. **Open the student notebook**
   in Google Colab (link in the README) — no install needed.

2. **Run the first setup cell**
   it installs umap-learn & adjustText and loads the data from GitHub.

3. **Fill in the blanks ___**
   in each ✏ Exercise cell, then run it.

4. **Stuck?**
   raise your hand — or peek at the solutions notebook.

Anatomy of a Matplotlib figure



Implicit (plt.)

```
plt.plot(x, y)
plt.title('...')
# which plot am I
# editing now?
```

Explicit / OO ✓ recommended

```
fig, ax = plt.subplots()
ax.plot(x, y)
ax.set_title('...')
ax.set_xlabel('...')
```

- **Figure** = the canvas (size, saving)
- **Axes** = one plot (data, labels, ticks)
- Seaborn draws *on* Axes → anything is customisable

Seaborn speaks tidy data

Wide (genes × samples)

gene	MW1	MW4	…
IL6	2.1	3.5	…
SOD2	7.3	10.4	…
C3	6.3	8.3	…



```
.melt()
.merge(meta)
```

Long / tidy (1 row = 1 observation)

gene	sample	log2cpm	tissue	condition
IL6	MW1	2.1	cornea	mock
IL6	MW4	3.5	cornea	CoV2
SOD2	MW1	7.3	cornea	mock
…	…	…	…	…

Axes-level

`sns.scatterplot` · `boxplot` · `violinplot` · `heatmap`

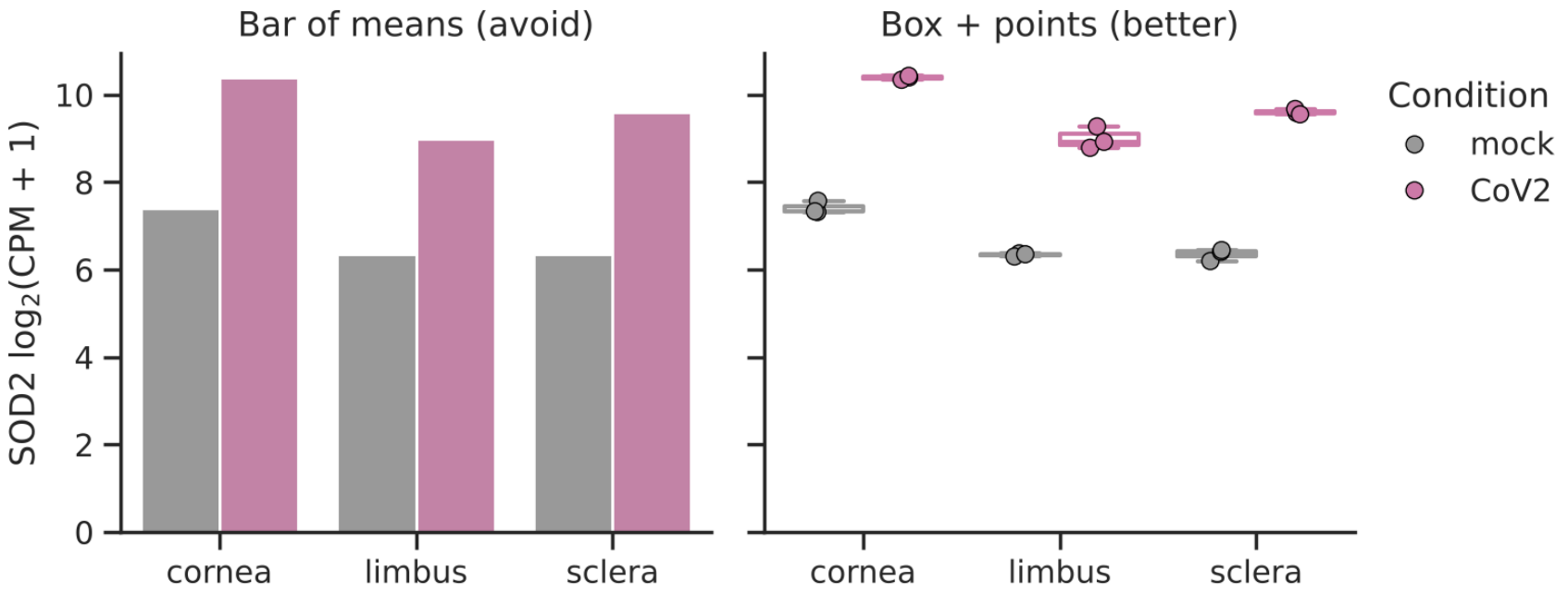
Draw on one Axes → pass ax= and place them in your own layout.

Figure-level

`sns.relplot` · `catplot` · `displot` · `clustermap`

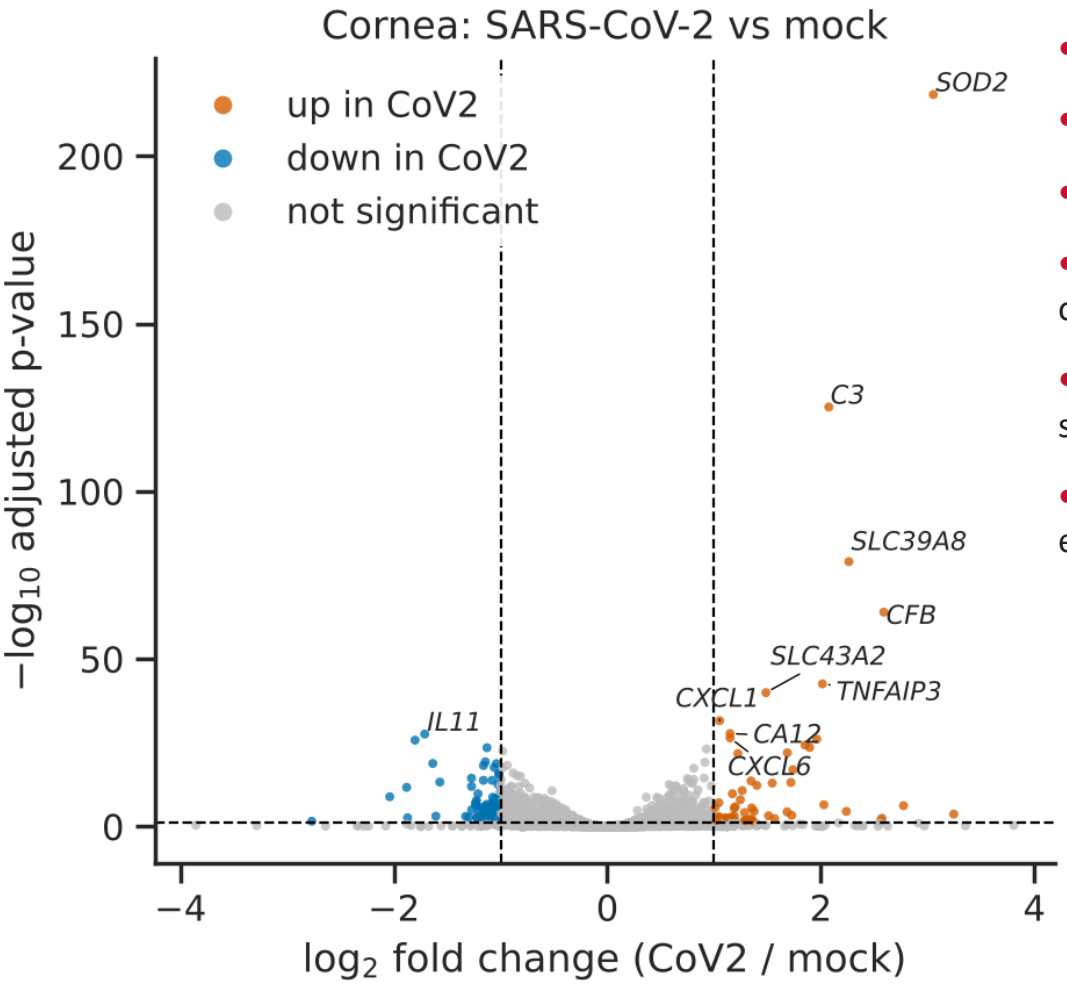
Create their own figure → easy small multiples with col= / row=.

With small n, show the points



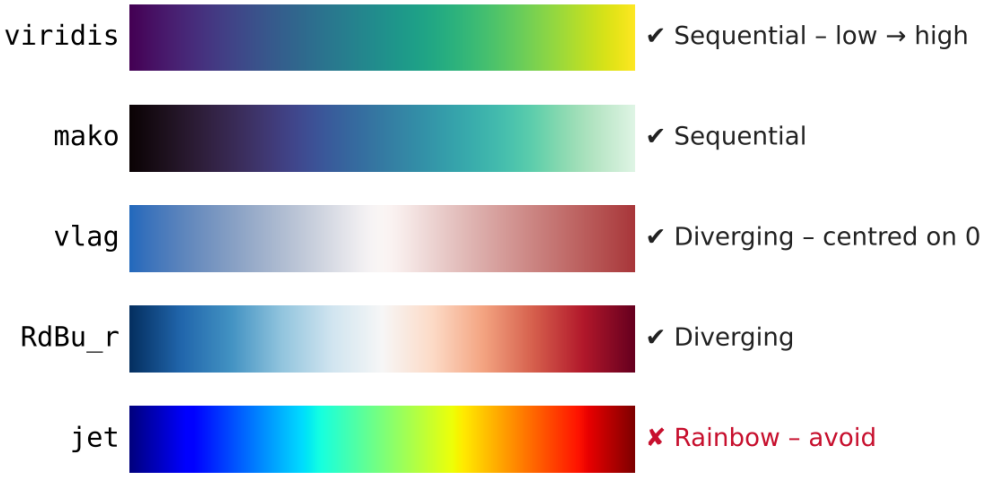
- **n = 3 per group:** a bar of means hides sample size, spread and outliers.
- **Overlay individual observations** (`sns.stripplot` on top of `sns.boxplot`) and define error bars in the legend.
- **Keep colours consistent:** define a palette dictionary once and reuse it everywhere.

Relationships: volcano & MA plots

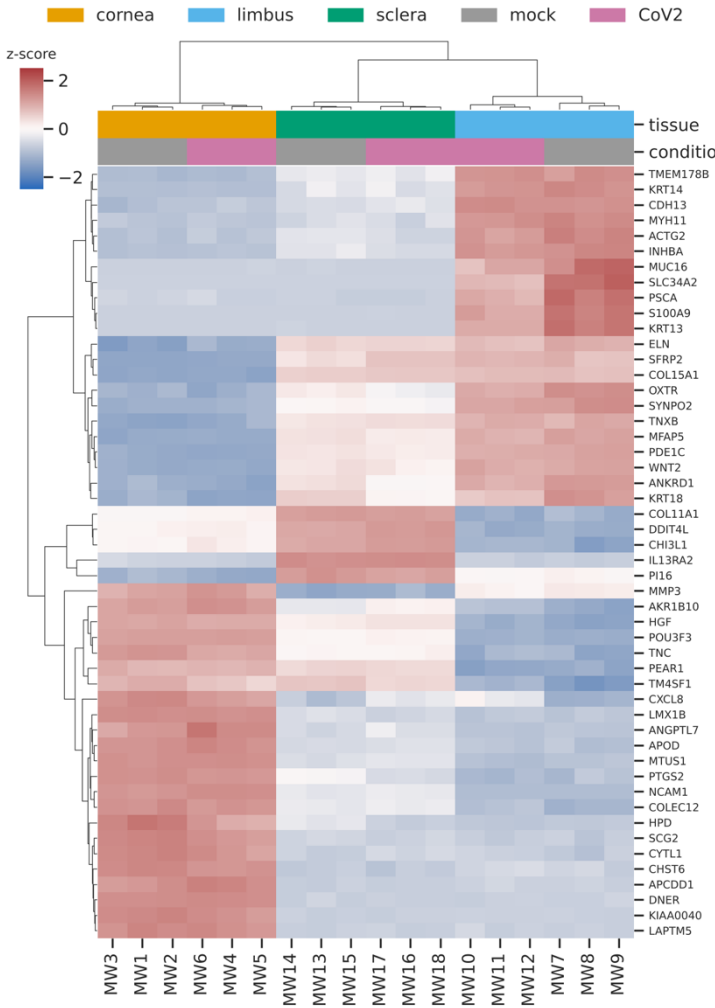


- **x**: effect size — $\log_2$ fold change
- **y**: significance — $-\log_{10}$ adj. p
- **Threshold lines** via ax.axhline / ax.axvline
- **Label key genes** — adjustText avoids overlaps
- **`rasterized=True`** for >10k points: small, fast vector files
- **Exercise 3:** build the MA plot (mean expression vs fold change)

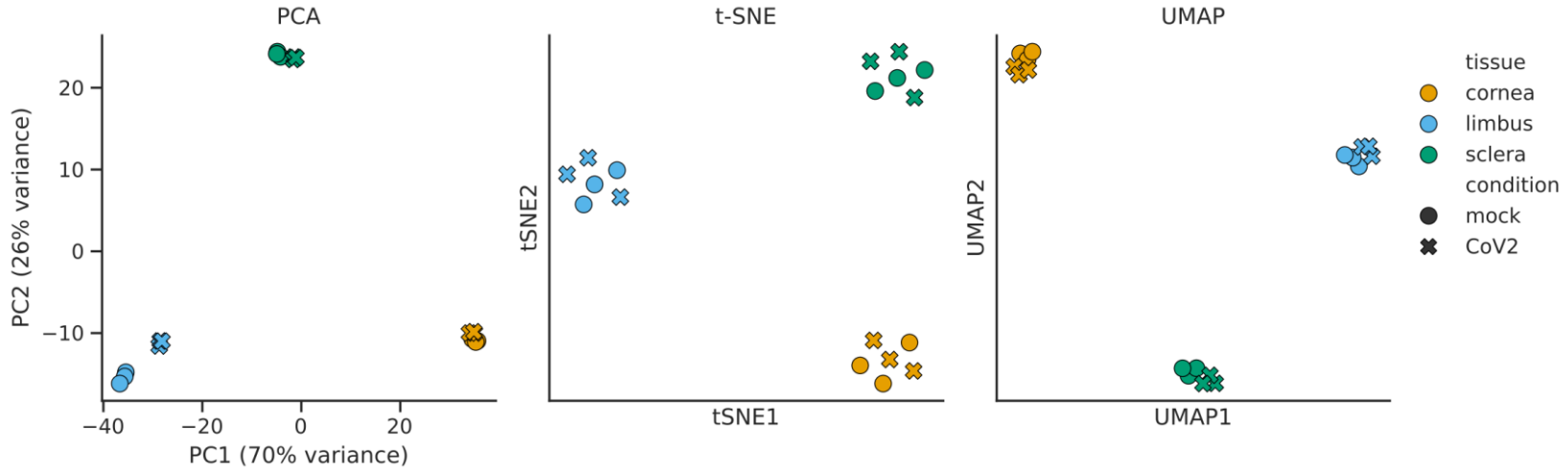
Heatmaps: choose the colour map on purpose



- **Sequential** for low → high (correlations, counts)
- **Diverging** + `center=0` for z-scores, fold changes
- **z-score genes** (`z_score=0`) to compare patterns
- **Annotate groups** with `col_colors`



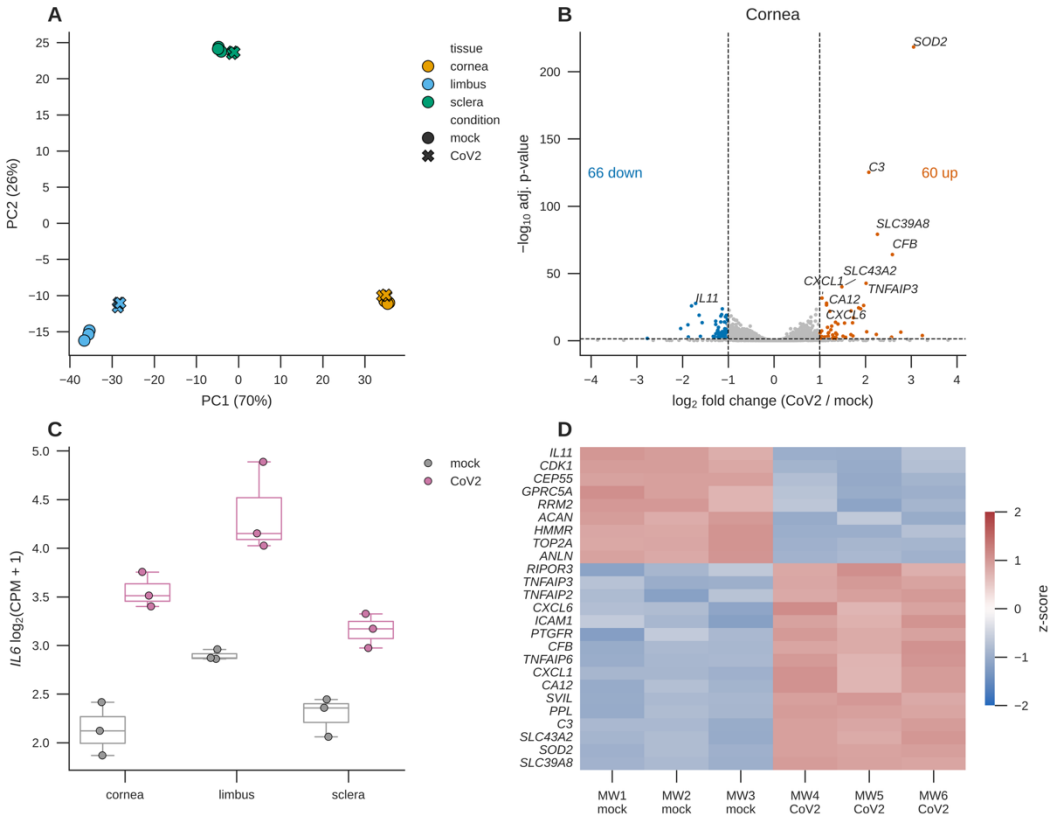
Dimensionality reduction at a glance



Tissue dominates PC1–PC2; infection only separates samples in a **within-tissue** PCA (notebook).

	PCA	t-SNE	UMAP
Type	linear	non-linear	non-linear
Axes mean…	% variance explained	nothing (no units)	nothing (no units)
Good for	bulk RNA-seq, global structure	local clusters	clusters + some global layout
Report	n features, scaling	perplexity, seed	n_neighbors, min_dist, seed

Multi-panel figures & export




```
MM = 1 / 25.4  # mm -> inch
fig, axd = plt.subplot_mosaic(
    'AB;CD',
    figsize=(180*MM, 140*MM),
    layout='constrained')

plt.rcParams['pdf.fonttype'] = 42
plt.rcParams['svg.fonttype'] = 'none'
# no bbox_inches='tight': keep 180 mm
fig.savefig('Fig1.pdf')
fig.savefig('Fig1.png', dpi=600)
```

- **Size in mm**: 85 (1 col) / 180 (2 col)
- **Fonts ≥ 6 pt** at final size
- **Editable text** in Inkscape via fonttype
- **Vector** for editing; **600 dpi** PNG

Panel D uses `sns.heatmap` with SciPy clustering — figure-level `clustermap` can't live inside a layout.

Publication-figure checklist

- ✓ **Labels**
  every axis labelled with units; gene names in italics
- ✓ **Colour**
  same meaning in every panel; colour-blind safe; no rainbow
- ✓ **Honesty**
  show individual points for small n; bars start at zero
- ✓ **Size**
  built at journal column width; fonts ≥ 6 pt when printed
- ✓ **Files**
  vector PDF/SVG + 600 dpi PNG; dense layers rasterised
- ✓ **Reproducible**
  code saved; seeds & parameters reported in Methods

What's next & resources

Coming up this week

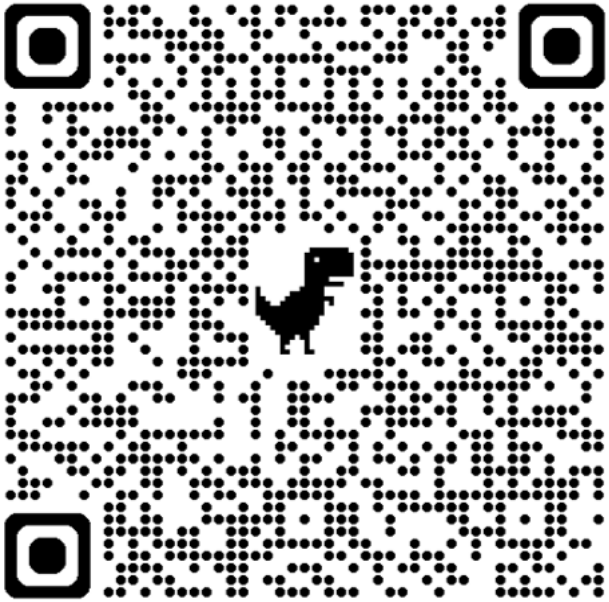
- **Day 2** · same plots in R with ggplot2
- **Day 3** · bring your own data — good & bad practices, hands-on
- **Day 4** · bring today's PDF/SVG to Inkscape
- **Day 5** · put the final figure into LaTeX

Learn more

- `matplotlib.org/cheatsheets`
- `seaborn.pydata.org/tutorial`
- `data-to-viz.com` — which chart for which data
- Rougier et al. 2014 *Ten simple rules for better figures*
- Crameri et al. 2020 *The misuse of colour in science communication*

Homework (5 min): keep your exported **`Figure1_cornea_CoV2.pdf / .svg`** — you will polish it in Inkscape on Thursday. Think about which of your own datasets you'd like to bring on Wednesday.

Feedback?



https://tinyurl.com/QLS-MiCM-Feedback